

A Note on the Integrity of Trees

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ABSTRACT. We amend some results in the literature. We verify a formula for the integrity of the binomial tree, and correct formulas for the integrity of the complete k -ary tree.

Dedicated to Henda Swart with the greatest thanks.

1 Introduction

The integrity of a graph was introduced in 1987 by Barefoot, Entringer and Swart [3]. They defined the *integrity* $I(G)$ of a graph G as

$$\min_{S \subset V} \{|S| + m(G - S)\},$$

where $m(G - S)$ denotes the maximum order of a component of $G - S$, and the minimum is taken over all noncomplete subsets S of the vertex set V of G . A set S which achieves the minimum is called an *I -set*. Results on the integrity of trees are to be found in Barefoot et al. [4], in [7], and elsewhere. For a survey of integrity, see Bagga et al. [2].

In this note, we verify a formula for the integrity of the binomial tree given by Alpay and Kirlangiç [1], and correct their formula for the integrity of the complete k -ary tree.

2 Binomial Trees

The authors of [1] claim a value for the integrity of the binomial trees. The value is correct, but the proof is incomplete. The proof considers only some subsets of vertices. For example, it seems to claim that the root vertex must be in the I -set, but this is not true: indeed, there is for each tree an I -set that does not include the root vertex.

There are several equivalent definitions of a *binomial tree*. One recursive definition is to define the tree B_0 as a single vertex, and then the

rooted tree B_{i+1} is obtained by taking one copy of each of B_0 through B_i , adding a root, and making the old roots the children of the new root. In particular, the tree B_i has 2^i vertices. An equivalent definition is based on the corona of a graph. Recall that the **corona** of a graph is obtained by adding a new leaf adjacent to each existing vertex. Then B_{i+1} is the corona of B_i . For a discussion of binomial trees, see the text [6].

Following the notation of [7], we define $D_x(G)$ to be the minimum number of vertices whose removal leaves the maximum component order at most x (so that, for example, $D_1(G)$ is the vertex cover number, and $D_N(G) = 0$ if N is at least the order). Several authors have defined such a parameter; for example, Boesch et al. [5] called it **component order connectivity**.

Lemma 1 *Let G' be the corona of G . For $x \geq 2$, $D_x(G') = D_{\lfloor x/2 \rfloor}(G)$.*

PROOF. Consider a set S of vertices of G' . If we replace every leaf in S by its neighbor in G' , we obtain a set S' such that $m(G' - S') \leq m(G' - S)$. Therefore, we may restrict attention to S a subset of $V(G)$. It then follows that $m(G' - S) = 2m(G - S)$ (provided $S \neq V(G)$). That is, if $m(G' - S) \leq x$, then $m(G - S) \leq x/2$, as required. QED

Now, since B_n has a perfect matching, it follows that $D_1(B_n) = 2^{n-1}$. Since B_n is the corona of B_{n-1} , by the above lemma we have that $D_2(B_n) = D_3(B_n) = D_1(B_{n-1}) = 2^{n-2}$. In general, it follows that for $1 \leq x < 2^n$,

$$D_x(B_n) = 2^{n-1-\lfloor \log_2 x \rfloor}.$$

Hence,

$$I(B_n) = \min_x \{x + D_x(B_n)\} = \min_x \{x + 2^{n-1-\lfloor \log_2 x \rfloor}\}.$$

Clearly the minimum is achieved at x a power of 2. That is,

$$I(B_n) = \min_c \{2^c + 2^{n-1-c}\}.$$

Some calculus shows that if n is odd, then $c = (n-1)/2$ is optimal; and if n is even, then $c = n/2$ or $n/2 - 1$ is optimal. We thus obtain:

Theorem 1 *The integrity of the binomial tree for $n \geq 1$ is given by*

$$I(B_n) = \begin{cases} 2^{(n+1)/2} & \text{if } n \text{ odd,} \\ 3 \cdot 2^{n/2-1} & \text{if } n \text{ even.} \end{cases}$$

3 Complete k -ary Trees

Using the notation of [1], we define H_n^k as the **rooted complete k -ary tree** of height $n - 1$: each vertex except the leaves has k children, and all leaves are distance $n - 1$ from the root. Thus H_n^k has order $(k^n - 1)/(k - 1)$.

The formulas given in [1] for the integrity of these trees are incorrect. For example, it is easily shown that $I(H_4^2) = 5$, but this is not the value from that paper. See Figure 1.

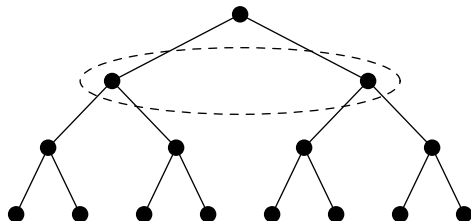


Figure 1: The complete binary tree H_4^2 (an I -set is circled)

We will show the following result:

Theorem 2 *For $k \geq 2$, the integrity of the complete k -ary tree of height $n - 1$ is given by*

$$I(H_n^k) = \begin{cases} \frac{k^{(n+1)/2} - 1}{k - 1} & \text{if } n \text{ is odd,} \\ \frac{(2k - 1)k^{n/2-1} - 1}{k - 1} & \text{if } n \text{ is even.} \end{cases}$$

As a special case we get the result for complete binary trees:

Corollary 1 *The integrity of the complete binary tree of height $n - 1$ is given by*

$$I(H_n^2) = \begin{cases} 2^{(n+1)/2} - 1 & \text{if } n \text{ is odd,} \\ 3 \cdot 2^{n/2-1} - 1 & \text{if } n \text{ is even.} \end{cases}$$

In order to prove the theorem, we first establish a property of I -sets of such trees.

3.1 Leveled trees

For a rooted tree, the i^{th} level $L(i)$ is the set of vertices at distance i from the root. A **leveled** tree is a rooted tree such that, for each i , every node

in $L(i)$ has the same number of children. We show that, as expected, every leveled tree has an I -set that is the union of levels.

We will need the following idea. Consider a graph G constructed as follows. Take r disjoint isomorphic graphs $F_1, \dots, F_r$ and another disjoint graph H . Say ϕ_i is an isomorphism from F_1 to F_i ($2 \leq i \leq r$). Let v be any vertex of F_1 and w any vertex of H , and add edges joining w to v and to each $\phi_i(v)$ ($2 \leq i \leq r$). See Figure 2.

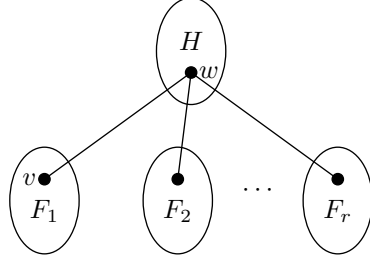


Figure 2: Forming G

We show next that, as expected, there is an I -set which is the ‘same’ on each F_i .

Lemma 2 *Let the graph G be constructed as above. Then there exists an I -set S such that $\phi_i(V(F_1) \cap S) = V(F_i) \cap S$ for $2 \leq i \leq r$.*

PROOF. Let S be any I -set. The result is clear if $w \in S$. So assume otherwise.

Consider first the case that not all $|S \cap V(F_i)|$ are equal. Without loss of generality, we may assume that $|S \cap V(F_1)|$ has the smallest value. Then, let S' be the set obtained from S by replacing $S \cap V(F_i)$ with the image $\phi_i(V(F_1) \cap S)$ for $2 \leq i \leq r$, and adding the cut-vertex w . It follows that $|S'| \leq |S|$ and $m(G - S') \leq m(G - S)$, so that S' is an I -set too. Further, S' has the desired property.

Consider second the case that all $|S \cap V(F_i)|$ are equal. Then let C be the component of $G - S$ containing w . Without loss of generality, we may assume that $|V(C) \cap V(F_1)|$ is the smallest value of $|V(C) \cap V(F_i)|$. So, let S'' be the set obtained from S by replacing $S \cap V(F_i)$ with the image $\phi_i(V(F_1) \cap S)$ for $2 \leq i \leq r$. It follows that $|S''| = |S|$ and $m(G - S'') \leq m(G - S)$, so that S'' is an I -set too. Further, S'' has the desired property, as required. QED

By repeatedly applying the idea in the previous proof at each vertex of the tree working upwards, we obtain the following:

Corollary 2 *For any leveled tree, there is an I -set that is the union of levels.*

3.2 Proof of Theorem 2

Let $G = H_n^k$. Consider first n being odd, and let $E = (k^{(n+1)/2} - 1)/(k - 1)$. Let $d = (n - 1)/2$ and let $S^* = L(d)$. Then $|S^*| = k^d$. Further, $G - S^*$ is $k^{d+1} + 1$ copies of H_d^k , and so $m(G - S^*) = (k^d - 1)/(k - 1)$. This shows that $I(H_n^k) \leq k^d + (k^d - 1)/(k - 1) = E$, and so the claimed value is an upper bound.

Now, by the above corollary, we have an I -set S that consists of levels. Since $|L(d+1)| = k^{d+1} > E$, S cannot contain $L(d+1)$ nor any bigger level. If S contains $L(d)$, then $m(G - S) = m(G - S^*)$ and so it must be that $S = S^*$. If S does not contain $L(d)$, then $m(G - S) \geq (k^{d+1} - 1)/(k - 1) = E$, and so S is not an I -set.

Consider next n being even, and let $E = ((2k - 1)k^{n/2-1} - 1)/(k - 1)$. Let $d = n/2 - 1$ and let $S^* = L(d)$. Then $|S^*| = k^d$. Further, $G - S^*$ contains k^{d+1} copies of H_{d+1}^k , and so $m(G - S^*) = (k^{d+1} - 1)/(k - 1)$. This shows that $I(H_n^k) \leq k^d + (k^{d+1} - 1)/(k - 1) = E$, and so the claimed value is an upper bound.

Now, by the above corollary, we have an I -set S that consists of levels. If $k \geq 3$, then $|L(d+1)| > E$, and so S cannot contain $L(d+1)$ nor any bigger level. If $k = 2$, then $|L(d+2)| > E$, and so S cannot contain $L(d+2)$ nor any bigger level; but if $L(d+1)$ is the largest level in S , then $|S| \geq 2^{d+1}$ and $m(G - S) \geq 2^d - 1$, so that $|S| + m(G - S) \geq E$.

Further, if S contains $L(d)$, then $m(G - S) = m(G - S^*)$, and so it must be that $S = S^*$. If S does not contain $L(d)$ nor any bigger level, then $m(G - S) \geq (k^{d+2} - 1)/(k - 1) > E$, and so S is not an I -set.

This establishes Theorem 2.

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